COURSE FEEDBACK SENTIMENT ANALYSIS USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled "COURSE FEEDBACK SENTIMENT ANALYSIS USING ZOHO CREATOR" is the bonafide work of I. Mohammed Hamza (2116240701326), who carried out the project work for the course CB23F35 – Low-Code Application Development using Zoho Creator under my supervision.

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Submitted for the Practical Examination held on _____.

DECLARATION

I, I. Mohammed Hamza (2116240701326), hereby declare that the project report titled "COURSE FEEDBACK SENTIMENT ANALYSIS USING ZOHO CREATOR", submitted for the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the work carried out by me. The Zoho Creator application described in this report, including its Course Feedback form, teacher and course management, feedback reports and administrator dashboard, was built by me.

I further declare that the sentiment classification in this application is performed by the built-in Zia Sentiment Analysis field of Zoho Creator, that no custom machine learning model was developed, and that every reference and external resource used in this work has been acknowledged in the report.

Place: _____

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I. Mohammed Hamza

ABSTRACT

Colleges collect course feedback from students so that teaching can be reviewed and improved, but the written comments are usually read one by one, if they are read at all, and it is hard to see how students feel about each teacher and course. This project presents Course Feedback Sentiment Analysis, a low-code application built on the Zoho Creator platform. Students submit feedback through a simple form with three fields: Teacher Name, Course / Subject Name and Course Feedback (multi-line text). The form has no sentiment dropdown. Instead, the application uses the native Sentiment Analysis field of Zoho Creator, a built-in Zia AI feature configured with Course Feedback as its source field, which automatically classifies every submission as Positive, Negative or Neutral. No custom machine learning model was developed or trained, and the classification does not rely on manually written keyword rules or Deluge scripts. An administrator dashboard shows the total number of feedback submissions, the Positive, Negative and Neutral counts, teacher-wise positive, negative and neutral statistics, sentiment distribution charts and teacher-wise comparison charts, and feedback reports can be filtered for closer reading. The application gives administrators a quick and consistent view of student opinion without requiring students to label their own comments or administrators to read every comment.

Keywords: Sentiment Analysis, Zia AI, Zoho Creator, Low-Code Development, Course Feedback, Student Feedback Analytics, Dashboard

PROJECT AT A GLANCE

Project Title	Course Feedback Sentiment Analysis using Zoho Creator
Application Domain	Education – Student Course Feedback Analytics
Student Name & Register No.	I. Mohammed Hamza – 2116240701326
Platform	Zoho Creator
AI Feature	Native Zia Sentiment Analysis field (source field: Course Feedback); no custom machine learning model
Forms	Feedback Submission, Feedback Review, Course Catalog, Faculty Profile and Student Profile
Reports	Feedback report with filters (by Sentiment, Teacher Name, Course / Subject Name)
Pages / Dashboards	1 (administrator dashboard with counts, teacher-wise statistics and charts)
Workflows / Deluge Scripts	Configured form workflows and native Zia classification; no custom Deluge sentiment script claimed
Integrations	None (out of scope)
Roles Defined	Student and Faculty custom permission sets; Administrator retains administrative access
Test Cases	13 (TC-01 to TC-13)
Sample Data	5 teachers, 5 courses and 6 feedback records
Client Used	Google Chrome (Version 157.0), Windows 10/11 laptop

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
CO	Course Outcome
FR	Functional Requirement
ID	Identifier
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
TC	Test Case
UC	Use Case
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Student feedback is an important source of information about teaching and courses in higher education. This project presents an academic course feedback system built in Zoho Creator. Students submit written comments, while faculty review feedback through a separately configured permission set. The application uses Zoho Creator's native Zia Sentiment Analysis to classify comments as Positive, Negative or Neutral and provides reports and a dashboard for monitoring feedback.

Written comments carry the most detail, but they are the hardest part of feedback to summarise. When many students comment on many teachers and courses, an administrator cannot easily tell how many comments are favourable, how many are critical, or which teachers receive more of one kind than the other. Sentiment analysis, a natural language processing task that classifies text by the opinion it expresses (Pang and Lee, 2008; Liu, 2012), can summarise such comments as Positive, Negative or Neutral.

Low-code platforms allow applications to be built mainly through visual configuration (Sahay et al., 2020). Zoho Creator is such a platform, and it provides a built-in Sentiment Analysis field powered by Zia, Zoho's AI assistant (Zoho Corporation, 2026a). This project uses that field to build a course feedback application in which the sentiment of every comment is generated automatically.

1.2 Problem Statement

Course feedback comments are written as free text and are difficult to review consistently as their volume increases. Institutions need a structured way to collect feedback, review and resolve it, and summarize student sentiment by course and faculty member. The system therefore combines student submission, faculty review, administrative monitoring and native Zia sentiment categorization. Access must also be limited by responsibility: Student and Faculty custom permission sets restrict access to the relevant submission and review forms, while the Administrator retains administrative access.

1.3 Objectives

- To build a simple feedback form that collects the Teacher Name, the Course / Subject Name and a multi-line Course Feedback from students.
- To use the built-in Zia Sentiment Analysis field of Zoho Creator, with Course Feedback as its source field, so that every submission is classified automatically as Positive, Negative or Neutral without any manual selection by the student.

- To maintain teacher and course data so that feedback is grouped consistently by teacher and course.
- To present the total number of submissions, the sentiment counts and teacher-wise sentiment statistics on an administrator dashboard, together with sentiment distribution and teacher-wise comparison charts.
- To provide feedback reports and a dashboard for monitoring submissions and sentiment trends.

1.4 Scope of the Project

In scope: student feedback submission; faculty and course management; feedback review and resolution; automatic sentiment classification using Zoho Creator's native Zia Sentiment Analysis; sentiment categorization as Positive, Negative or Neutral; reports and dashboard monitoring; form workflows; and role-based access through Student and Faculty custom permission sets and Administrator access.

Out of scope: developing or training a custom machine learning model, rule-based or keyword-based sentiment classification, analysis of text that is not in English, numeric rating scales, aspect-level analysis of individual topics within a comment, and integration with other systems.

1.5 Role of AI and Justification for a Low-Code Approach

The project uses artificial intelligence in one place: the Sentiment Analysis field of Zoho Creator. This is a ready-made Zia AI field provided by the platform. In the Course Feedback form it is configured with the Course Feedback field as its source field. When a feedback record is saved, the built-in AI feature analyses the text of Course Feedback and stores the result, Positive, Negative or Neutral, in the sentiment field of the same record.

The sentiment is therefore generated automatically by the built-in AI feature. It is not chosen by the student from a dropdown, and it is not produced by keyword lists, conditions or Deluge scripts written for this project. No custom machine learning model was developed, trained or deployed. According to the Zoho Creator documentation, the field is estimated by the platform's algorithm and appears disabled on the form so that users cannot enter a value, and sentiment analysis can currently be performed on English text (Zoho Corporation, 2026a).

Building a sentiment classifier from scratch needs labelled text, model training and evaluation, which is outside the aim of a low-code project. Using the built-in field avoids training a model, preparing labelled data and calling an external service from code, which suits a small institutional application. Zoho Creator also provides forms, reports, dashboards and AI fields on one hosted platform (Zoho Corporation, 2026a), so

the feedback form, the sentiment field and the administrator dashboard can all be built and hosted in one place.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the problem definition, requirements, use cases and user access needs. Chapter 4 describes the architecture, data model, process and permission design. Chapter 5 details the forms, Zia configuration, reports, dashboard, workflows and User Permissions configuration. Chapter 6 presents functional and RBAC testing. Chapter 7 discusses deployment, security and governance. Chapter 8 presents the results, and Chapter 9 concludes with future enhancements.

Throughout the report, the built-in AI feature is described as it is used in the application, and statements about its limits are taken from the Zoho Creator documentation and from the results of this project.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical manual feedback process, course feedback is collected on paper forms or on general-purpose online forms. The written comments are then read one by one. The sentiment of a comment is judged by the person reading it, or it is chosen by the student from a list. A teacher-wise summary is compiled by hand, charts are built manually if at all, and the records are sorted manually when an administrator wants to look at a particular teacher or kind of comment.

Every step of this process depends on a person reading, judging and counting, so the result varies with the reader and takes longer as the number of comments grows.

2.2 Limitations of the Existing System

- Comments are read one by one, which is slow and inconsistent as the number of comments grows.
- When the student chooses the sentiment label, it depends on the student's choice rather than on the words written, and it adds another field to fill in.
- The sentiment judged by reading depends on the person reading the comment, so the labelling is not consistent.
- Teacher-wise summaries of positive, negative and neutral comments are compiled by hand.
- Administrators lack a quick view of how many comments are positive, negative or neutral, both overall and for each teacher.
- Charts are built manually, if at all, and records are sorted manually.

2.3 Custom Model versus Built-in Zia Field

Table 2.1 compares building a sentiment classifier from scratch with using the built-in Sentiment Analysis field of Zoho Creator for this application. The comparison is based on the requirements described in the literature and in the Zoho Creator documentation; no effort or time measurements were taken.

Table 2.1: Comparison of a Custom Model and the Built-in Zia Field for the Proposed Application

Criterion	Custom Sentiment Model	Built-in Zia Sentiment Analysis Field
Labelled data	Labelled text is needed	Not needed; the platform provides the feature
Model training and evaluation	Model has to be trained and evaluated	No model is developed, trained or deployed in this project
Calling a service from code	An external service has to be called from code if one is used	Not needed; the field is added through the form builder
Source of the label	Output of the model that was built	Estimated by the platform's algorithm from the source field
Entry by the student	Depends on how the form is designed	Field appears disabled; a user cannot enter a value
Language	Depends on the model	Currently English text (Zoho Corporation, 2026a)
Fit for this project	Outside the aim of a low-code project	Suits a small institutional application

2.4 Literature Survey

The review covers four areas: student feedback in higher education, sentiment analysis, the Sentiment Analysis field of Zoho Creator, and low-code development platforms. Table 2.2 summarises the sources studied.

Student feedback. Marsh (2007) reviewed research on students' evaluations of university teaching, including its reliability, validity, potential biases and usefulness. Spooren, Brockx and Mortelmans (2013) reviewed the state of the art on the validity of student evaluation of teaching. Both works treat student evaluations as a source of information that must be interpreted with care, which supports collecting feedback from many students and viewing it in summary form instead of relying on a single comment.

Sentiment analysis. Sentiment analysis, also called opinion mining, is the computational study of the opinions and sentiments expressed in text. Pang and Lee (2008) survey its techniques and applications, and Liu (2012) describes the task at the document, sentence and aspect levels. Medhat, Hassan and Korashy (2014) survey sentiment analysis algorithms and applications, and Zhang, Wang and Liu (2018) survey deep learning approaches to the task.

Low-code platforms. Sahay et al. (2020) compared low-code development platforms, which let applications be built mainly through visual configuration instead of hand-written code.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Marsh, 2007	Students' Evaluations of University Teaching: Dimensionality, Reliability, Validity, Potential Biases and Usefulness	Reviews the reliability, validity, potential biases and usefulness of student evaluations of teaching.	Motivates collecting student feedback and summarising it carefully.
2	Spooren, Brockx and Mortelmans, 2013	On the Validity of Student Evaluation of Teaching: The State of the Art	Reviews the state of the art on the validity of student evaluation of teaching.	Motivates collecting student feedback and summarising it carefully.
3	Pang and Lee, 2008	Opinion Mining and Sentiment Analysis	Surveys the techniques and applications of sentiment analysis (opinion mining).	Defines the task performed by the AI field.
4	Liu, 2012	Sentiment Analysis and Opinion Mining	Describes sentiment analysis at the document, sentence and aspect levels.	Defines the task performed by the AI field.
5	Medhat, Hassan and Korashy, 2014	Sentiment Analysis Algorithms and Applications: A Survey	Surveys sentiment analysis algorithms and applications.	Shows the task is well established; supports using a ready-made service instead of building a model.
6	Zhang, Wang and Liu, 2018	Deep Learning for Sentiment Analysis: A Survey	Surveys deep learning approaches to sentiment analysis.	Shows the task is well established; supports using a ready-made service instead of building a model.
7	Sahay et al., 2020	Supporting the Understanding and Comparison of Low-Code Development Platforms	Compares low-code development platforms.	Choice of Zoho Creator as the platform.
8	Zoho Corporation, 2026a	Understand Sentiment Analysis Field (Zoho Creator Help)	The Sentiment Analysis field labels text as Positive, Neutral or Negative; English text is currently supported.	The AI component of this project.
9	Zoho Corporation, 2026b	Add Sentiment Analysis Field to a Form (Zoho Creator Help)	The field is added through the form builder by selecting the source field.	Used to configure the Sentiment Analysis field.

2.5 Sentiment Analysis Field in Zoho Creator

Zoho Creator provides AI fields that can be added to a form. The Sentiment Analysis field determines whether the text in a single-line or multi-line field is Positive, Neutral or Negative (Zoho Corporation, 2026a). The source field must be present in the same form. Because the value is estimated by the algorithm, the field appears disabled on the form, which tells users that they cannot enter a value. The field can also be shown only to the application's administrator, and sentiment analysis can currently be performed on English text. The field is added through the form builder by selecting the source field (Zoho Corporation, 2026b).

Using this built-in field avoids training a model, preparing labelled data and calling an external service from code, which suits a small institutional application.

2.6 Proposed System

Course Feedback Sentiment Analysis is a Zoho Creator application in which every student submission becomes a Course Feedback record containing the Teacher Name, the Course / Subject Name and the Course Feedback text. A Sentiment Analysis field, whose source field is Course Feedback, automatically assigns Positive, Negative or Neutral to the record. The administrator dashboard and reports then summarise the stored records.

Each limitation in Section 2.2 is addressed as follows:

- One simple form with three fields replaces unstructured collection of feedback.
- The Zia Sentiment Analysis field labels every comment automatically, so no student has to select a sentiment and nobody has to read every comment in order to classify it.
- The same built-in AI feature labels every record, so the labelling does not depend on who reads the comment.
- Teacher-wise statistics show the positive, negative and neutral counts for each teacher at a glance.
- Dashboard charts and filterable reports let administrators move from the summary to the individual comments.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The application has three user categories: Student, Faculty and Administrator. Student and Faculty are custom permission sets configured through Zoho Creator User Permissions; they are not custom roles in the role hierarchy. The default role hierarchy contains the CEO role. Students submit course feedback and are restricted from the Feedback Review form. Faculty access the Feedback Review form and are restricted from the Feedback Submission form. The Administrator retains administrative access. Other component access is limited to the permissions configured for each user and is not assumed here.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Student	Submits course feedback; Student custom permission set allows Feedback Submission and restricts Feedback Review.	Uses permitted submission components only; other access is as configured.
Administrator	Retains administrative access; oversees application data and monitoring.	Administrative access.
Teacher / Course (data)	Entries against which feedback is recorded.	Stored as records; they do not use the application.
Faculty (custom permission set)	Reviews and resolves feedback through Feedback Review; Faculty set restricts Feedback Submission.	Access to other components is as configured.

3.2 Functional Requirements

The functional requirements below are derived from the objectives in Section 1.3 and the modules described in Chapter 4.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Module
FR-01	The system shall provide a form in which a student enters the Teacher Name, the Course / Subject Name and a multi-line Course Feedback.	Student	1
FR-02	The system shall not offer a sentiment dropdown or rating selector to the student; the Sentiment field shall appear disabled.	System	1

FR-03	The system shall classify every saved feedback record as Positive, Negative or Neutral using the Zia Sentiment Analysis field with Course Feedback as its source field.	System	3
FR-04	The system shall store each feedback record together with the generated sentiment.	System	1, 3
FR-05	The system shall allow the administrator to add, edit and remove teacher and course entries.	Administrator	2
FR-06	The system shall show the positive, negative and neutral feedback counts for each teacher.	Administrator	4
FR-07	The system shall show the total number of submissions and the Positive, Negative and Neutral counts on a dashboard.	Administrator	5
FR ID	**Requirement**	**Role(s)**	**Module**
FR-08	The system shall show sentiment distribution charts and teacher-wise comparison charts.	Administrator	5
FR-09	The system shall provide feedback reports showing Teacher Name, Course / Subject Name, Course Feedback and Sentiment, which can be filtered.	Administrator	6
FR-10	The system shall restrict the Student permission set from accessing Feedback Review while allowing Feedback Submission.	Student	User Permissions
FR-11	The system shall restrict the Faculty permission set from accessing Feedback Submission while allowing Feedback Review.	Faculty	User Permissions
FR-12	The Administrator shall retain administrative access to the application.	Administrator	User Permissions

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	The feedback form shall be simple, with three fields for the student to fill in, and shall be usable from a standard web browser.
NFR-02	Consistency	The same built-in AI feature shall label every record, so that the labelling does not depend on who reads the comment.
NFR-03	Data integrity	Teacher and course data shall be maintained so that the same names are used in every feedback record and the teacher-wise statistics group correctly.
NFR-04	Language	Sentiment analysis currently works on English text (Zoho Corporation, 2026a); feedback in other languages is outside the scope.

NFR-05	Accuracy	The label is an estimate made by the platform; its accuracy was not independently evaluated in this project.
NFR-06	Maintainability	The classification relies on a built-in platform feature; no custom model, keyword rule or Deluge script has to be maintained.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
AI service	Native Zia Sentiment Analysis field of Zoho Creator (source field: Course Feedback); no custom machine learning model
Application components	Course Feedback form, teacher and course management, feedback reports, administrator dashboard
Browser	Google Chrome (Version 157.0)
Client hardware	Windows 10/11 laptop
Sample data	5 teachers, 5 courses and 6 feedback records entered for testing

3.5 Use Case Model

Figure 3.1 shows the principal use cases. Students submit feedback; Zoho Creator's native Zia field classifies the saved comment. Faculty review and resolve feedback using the Feedback Review form. Administrators retain application administration and monitor feedback through reports and the dashboard. Student and Faculty form access is restricted through their respective custom permission sets.

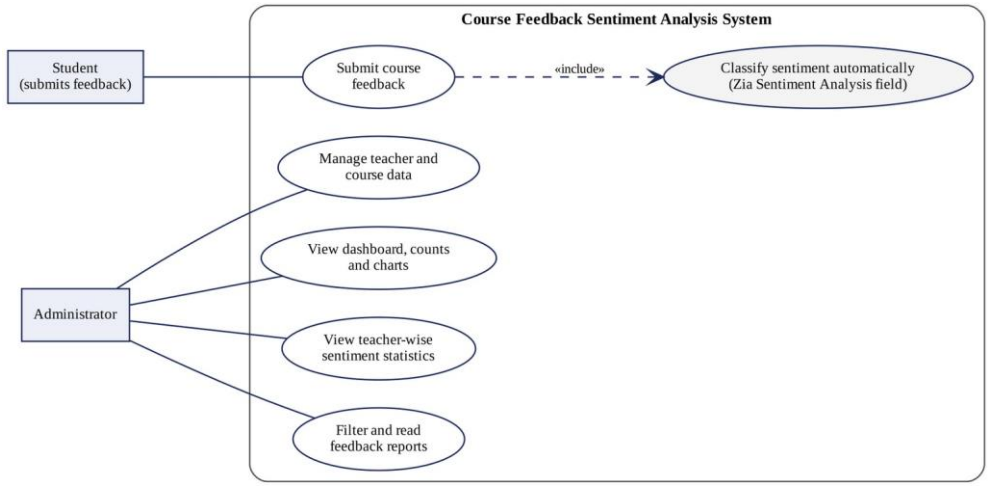


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit Course Feedback

Item	Description
Use Case ID	UC-01
Use Case Name	Submit course feedback
Actor(s)	Student
Pre-condition	Teacher and course entries exist so that they can be selected in the form.
Main Flow	1. The student opens the Course Feedback form. 2. The student enters the Teacher Name, the Course / Subject Name and the Course Feedback (multi-line text). 3. The student submits the form; the feedback record is saved in the Zoho Creator database. 4. The built-in Zia Sentiment Analysis field classifies the Course Feedback text as Positive, Negative or Neutral. 5. The sentiment is stored with the record, and the dashboard and reports include it.
Alternate Flow	If a required field is empty, the record is not saved and a message asks for the missing field (TC-03).
Post-condition	The feedback record exists with an automatically generated sentiment.
Related FR	FR-01, FR-02, FR-03, FR-04

Table 3.6: Use Case Description – Review Feedback Analytics

Item	Description
Use Case ID	UC-02
Use Case Name	Review feedback analytics
Actor(s)	Administrator
Pre-condition	Feedback records with generated sentiments are stored.
Main Flow	1. The administrator opens the dashboard and reads the total number of submissions and the Positive, Negative and Neutral counts. 2. The administrator reads the teacher-wise statistics and the sentiment distribution and teacher-wise comparison charts. 3. The administrator opens the feedback report and filters it, for example by Sentiment = Negative or by Teacher Name. 4. The administrator reads the comments behind the figures.
Alternate Flow	If no record matches the filter, no record is listed.
Post-condition	The administrator has reviewed the summary and the individual comments.
Related FR	FR-05, FR-06, FR-07, FR-08, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the architecture. Students use the Feedback Submission form, Faculty use the Feedback Review form, and administrators monitor permitted application components through a web browser. Zoho Creator User Permissions control access to the forms, while the native Zia field classifies feedback.

Figure 4.1: Architecture Diagram of the Course Feedback Sentiment Analysis System

Users and access channel. Students submit feedback, Faculty review feedback, and the Administrator monitors and administers the application through a web browser. Custom permission sets restrict form access.

Application layer. Zoho Creator hosts the Feedback Submission and Feedback Review forms, Course Catalog, Faculty Profile and Student Profile components, feedback reports, workflows and dashboard.

AI service layer. The Sentiment Analysis field is a built-in Zia AI field of Zoho Creator on the Course Feedback form, with Course Feedback as its source field. It generates Positive, Negative or Neutral automatically. No custom machine learning model, keyword rules or Deluge scripts are used for the classification.

Data layer. Teacher, Course and Course Feedback records, including the generated sentiment, are stored in the Zoho Creator database. The reports and the dashboard read these records to produce lists, counts and charts.

4.2 Data Model

Figure 4.2 shows the data model. A teacher can receive many feedback records and a course can be the subject of many feedback records. Each Course Feedback record

stores the Teacher Name, the Course / Subject Name, the Course Feedback text and the Sentiment value generated by the Zia field.

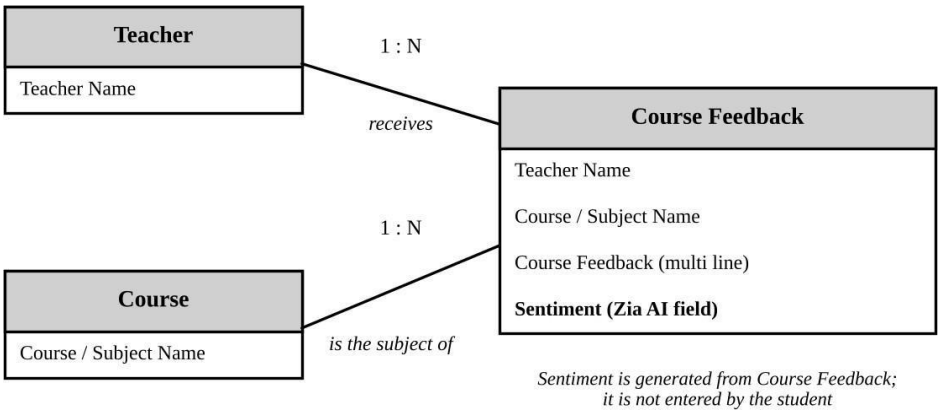


Figure 4.2: Data Model of the Application

Table 4.1: Entities and Relationships

Entity	Key Fields	Related To	Relationship Type	Implemented Using
Teacher	Teacher Name	Course Feedback	One-to-Many	Teacher Name dropdown in the Course Feedback form
Course	Course / Subject Name	Course Feedback	One-to-Many	Course / Subject Name dropdown in the Course Feedback form
Course Feedback	Teacher Name, Course / Subject Name, Course Feedback, Sentiment	Teacher; Course	Many-to-One	Dropdown fields; Sentiment is generated by the Zia field and not entered by the student

4.3 Form Design

The Course Feedback form has three fields that the student fills in: Teacher Name, Course / Subject Name and Course Feedback, a multi-line text field for the written comment. The form deliberately has no sentiment dropdown or rating selector, because the sentiment is not chosen by the student. When the form is submitted, the entries are stored as one Course Feedback record in the Zoho Creator database. The Sentiment Analysis field belongs to the same form; Zoho Creator shows this field as disabled so that a user cannot enter a value in it.

Table 4.2: Field Specification – Course Feedback Form

Field	Field Type	Filled By	Purpose
Teacher Name	Dropdown	Student	Identifies the teacher being reviewed
Course / Subject Name	Dropdown	Student	Identifies the course or subject
Course Feedback	Multi line	Student	Written feedback; source field of the sentiment field
Sentiment	Sentiment Analysis (Zia AI field)	Generated automatically	Positive, Negative or Neutral

4.4 Process and Workflow Design

Figure 4.3 shows the flow of one feedback submission. The student opens the form, enters the three fields and submits. The record is saved, and the built-in Zia Sentiment Analysis field classifies the Course Feedback text as Positive, Negative or Neutral. The sentiment is stored with the record, and the dashboard and reports then include the new record in their counts, statistics and charts. The classification is part of the platform's AI field and is not a script or a workflow written for this project.

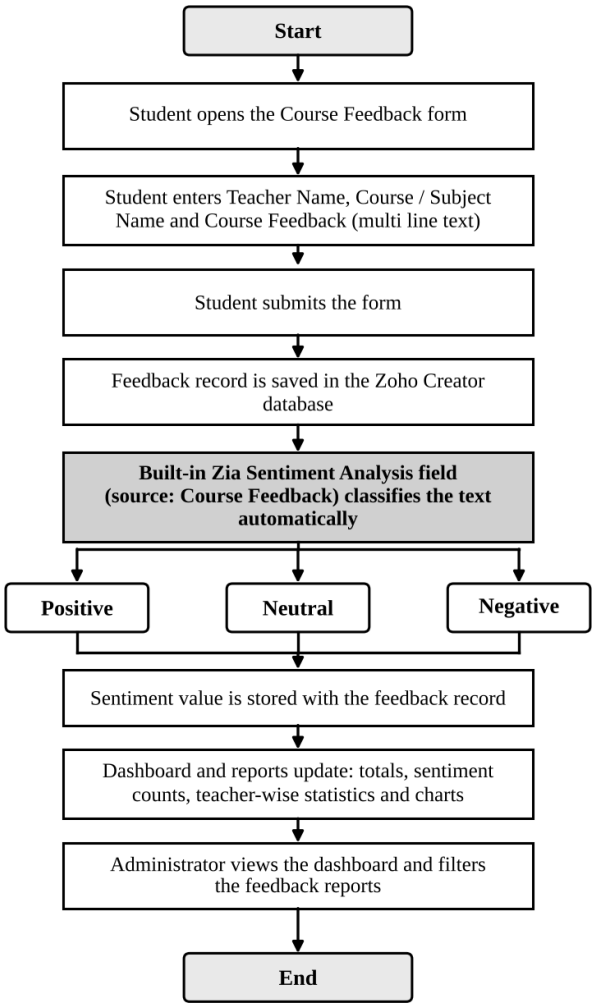


Figure 4.3: Process Flow of Course Feedback Submission and Sentiment Analysis

4.5 User Interface and Navigation Design

The application is used through a web browser. As seen in Figure 5.4, the application menu has the tabs Dashboard, Course Catalogs, Teacher Profiles and Feedback Submissions. The administrator dashboard summarises all stored feedback in one view. It shows the overall counts, the teacher-wise statistics and two kinds of charts, as listed in Table 4.3.

Table 4.3: Components of the Administrator Dashboard

Component	Information Displayed
Total feedback submissions	Number of Course Feedback records submitted
Positive, Negative and Neutral counts	Number of records with each AI-generated sentiment
Teacher-wise statistics	Positive, negative and neutral feedback counts for each teacher
Sentiment distribution charts	Share of Positive, Negative and Neutral feedback
Teacher-wise comparison charts	Comparison of sentiment counts across teachers

For every teacher, the application counts the feedback records whose sentiment is Positive, Negative and Neutral. The counts come directly from the stored Teacher Name and the AI-generated Sentiment fields, so no separate classification is performed for the statistics. Table 4.4 lists the statistics.

Table 4.4: Teacher-wise Sentiment Statistics

Statistic	Definition	Fields Used
Teacher-wise positive feedback	Number of feedback records of a teacher whose Sentiment is Positive	Teacher Name, Sentiment
Teacher-wise negative feedback	Number of feedback records of a teacher whose Sentiment is Negative	Teacher Name, Sentiment
Teacher-wise neutral feedback	Number of feedback records of a teacher whose Sentiment is Neutral	Teacher Name, Sentiment

4.6 Access Control Design

The system has Student and Faculty custom permission sets configured through Zoho Creator User Permissions, plus the Administrator's administrative access. The two test accounts confirmed the form restrictions: Student can access Feedback Submission and is restricted from Feedback Review; Faculty can access Feedback Review and is restricted from Feedback Submission. The permission matrix in Table 4.5 records only these confirmed restrictions and administrator access. Access to other components follows the configured permissions and is not generalized beyond what was verified.

Table 4.5: User Permission Matrix

Component	**Student**	**Faculty**	Administrator
Feedback Submission	Allowed	Restricted	Allowed
Feedback Review	Restricted	Allowed	Allowed
Course Management	As configured	As configured	Administrative access
Faculty Management	Restricted	As configured	Administrative access
Dashboard	As configured	As configured	Administrative access
Sentiment field	Generated by Zia; not editable	As configured	Administrative access

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application, Course Feedback Sentiment Analysis, was built on the Zoho Creator platform. Table 5.1 lists the software and hardware used. The AI component is the native Zia Sentiment Analysis field of Zoho Creator with Course Feedback as its source field; no custom machine learning model was used.

Table 5.1: Software and Hardware Used

Item	Specification
Platform	Zoho Creator
AI service	Native Zia Sentiment Analysis field of Zoho Creator (source field: Course Feedback); no custom machine learning model
Application components	Course Feedback form, teacher and course management, feedback reports, administrator dashboard
Client	Google Chrome (Version 157.0), Windows 10/11 laptop
Sample data	5 teachers, 5 courses and 6 feedback records entered for testing

The development steps were as follows:

1. A Zoho Creator application was created.
2. The Course Feedback form was built with the fields Teacher Name, Course / Subject Name and Course Feedback.
3. A Sentiment Analysis field was added to the form with Course Feedback as its source field, and its properties were checked.
4. Teacher and course management was set up.
5. Feedback reports with filters were configured.
6. The administrator dashboard was configured with the counts, teacher-wise statistics and charts.
7. Sample feedback was entered to test the application.

5.2 Forms and Data Capture

The Student Feedback Submission form collects the Teacher Name and Course / Subject Name and the written comment in the multi-line Course Feedback field. The form's native Zia Sentiment Analysis field uses Course Feedback as its source; Students cannot enter or select the generated sentiment. Faculty use the separate Feedback Review form to review and resolve feedback. The Course Catalog, Faculty Profile and Student Profile support the application's management components, with access governed by the configured User Permissions. Figures 5.1 and 5.2 show the

submission form and Zia configuration.

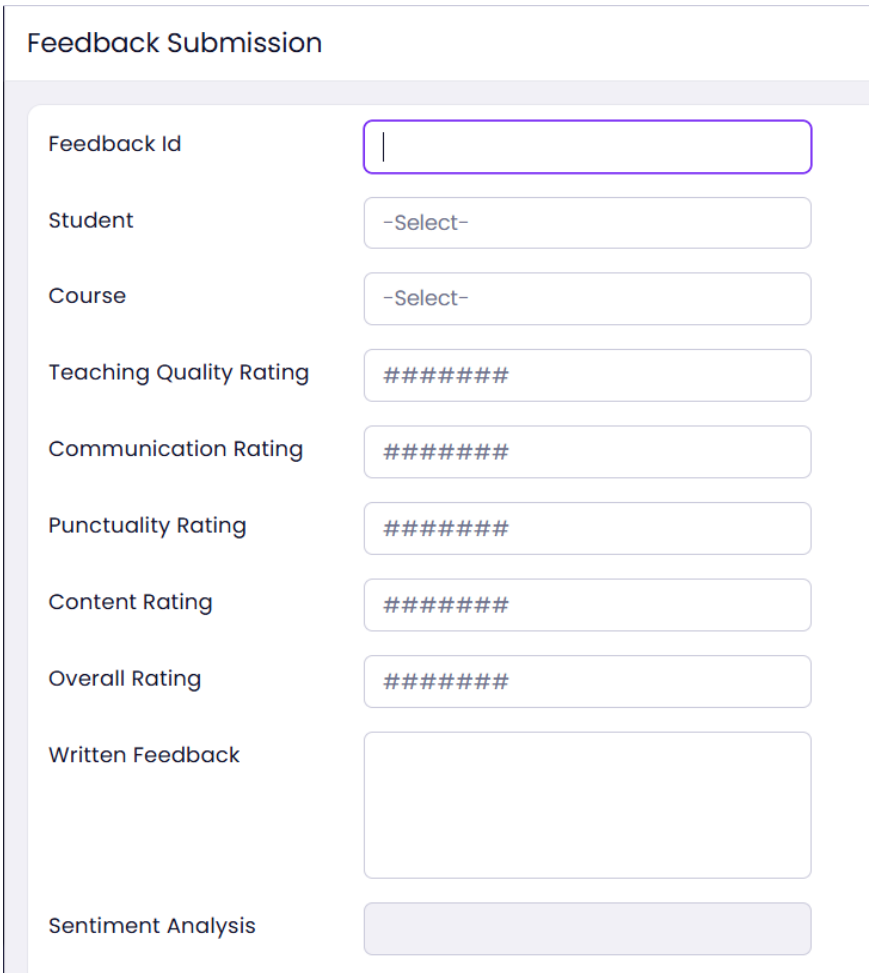


Figure 5.1: Course Feedback Form in Live Mode

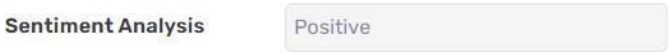


Figure 5.2: Zia Sentiment Analysis Field Configuration (Source: Course Feedback)

Teacher and course management maintains the teachers and the courses or subjects for which feedback is collected, so that the same names are used in every feedback record and the teacher-wise statistics group correctly. The administrator adds, edits and removes teacher and course entries.

5.3 Reports

The feedback report shows the stored Course Feedback records, with the Teacher Name, the Course / Subject Name, the Course Feedback text and the AI-generated Sentiment. The report can be filtered to narrow the records, for example to see only the Negative feedback of one teacher. This lets the administrator read the comments behind the figures shown on the dashboard. Figure 5.3 shows the feedback records with the AI-generated sentiment.

Table 5.2: Reports Implemented

Report Name	Type	Source Form	Filter / Use

Feedback report (All Feedback Submissions)	List	Course Feedback	Filter by Sentiment, Teacher Name or Course / Subject Name to read the comments behind the dashboard figures

All Feedback Submissions

Content Rating	Overall Rating	Written Feedback	Submission Date	Sentiment Category	Feedback Status
3	3	Good guidance	09-Nov-2026 18:16:24	Neutral	
3	3	Very disappointed with the lack of interaction today.	09-Nov-2026 14:20:00	Neutral	Submitted
5	5	Outstanding teaching quality and perfect punctuality.	09-Nov-2026 13:45:00	Negative	Resolved
5	4	Excellent material but communication could be improved.	09-Nov-2026 12:00:00	Neutral	Action Taken
3	4	Great content but class started late today.	09-Nov-2026 11:15:00	Neutral	Under Review
4	4	Instructor was very engaging and clear in explanations.	09-Nov-2026 10:30:00	Negative	Action Taken

Figure 5.3: Feedback Records with AI-Generated Sentiment

5.4 Pages and Dashboards

The dashboard supports feedback monitoring through submission totals, Positive, Negative and Neutral counts, teacher-wise statistics, sentiment distribution and comparison charts where configured. Feedback reports allow permitted users to inspect and filter stored comments. Administrator monitoring access is retained; other dashboard access remains as configured in User Permissions.

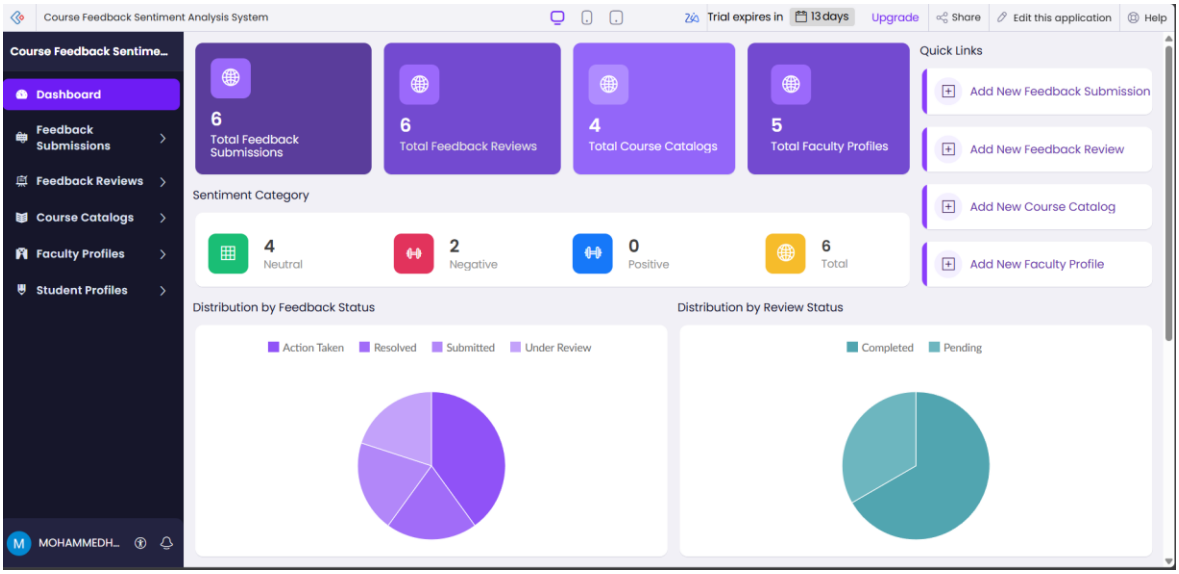


Figure 5.4: Administrator Dashboard: Submission and Sentiment Counts



Figure 5.5: Sentiment Distribution Chart.

5.5 Deluge Scripting

Sentiment classification is performed by Zoho Creator's native Zia Sentiment Analysis field, configured with the feedback text as its source. No custom Deluge sentiment-classification script, keyword rules, Python model, machine-learning model or external AI API is implemented. The available implementation evidence supports the configured form workflow and platform automation; no additional custom Deluge logic is claimed.

5.6 Workflows and Automation

The application includes form workflows for its submission and review processes. The available implementation details do not identify individual event triggers, conditions or Deluge actions, so this report does not specify or claim those details. Separately, when a feedback record is saved, Zoho Creator's native Zia Sentiment Analysis field analyses Course Feedback and stores Positive, Negative or Neutral in the same record. Reports and dashboard views use the stored record.

5.7 Integrations

Integration with other systems is out of scope for this project (Section 1.4). The application uses only the built-in Zia Sentiment Analysis field of Zoho Creator; no external service is called from code.

5.8 User Permissions and Role-Based Access

Zoho Creator User Permissions are used to assign the Student and Faculty custom permission sets to separate test users. The Student permission set grants access to Feedback Submission and restricts Feedback Review. The Faculty permission set grants access to Feedback Review and restricts Feedback Submission. These are custom permission sets, not custom roles in the role hierarchy; the default hierarchy contains the CEO role. The Administrator retains administrative access. Testing of the two accounts confirmed the restrictions (Table 6.1).

5.9 AI and Analytics Features

AI component: Zia Sentiment Analysis field. The Course Feedback form contains a field of type Sentiment Analysis, one of the Zia AI fields provided by Zoho Creator. Its source field is Course Feedback. When a record is saved, the built-in AI feature analyses the text in the source field and stores the result, Positive, Negative or Neutral, in the sentiment field.

This is a ready-made platform feature. No custom machine learning model was developed, trained or deployed for this project. The classification is not produced by keyword rules, conditions or Deluge scripts, and the student does not select it. According to the Zoho Creator documentation, the source field must be a single-line or multi-line field in the same form, the value is estimated by the algorithm, and sentiment analysis can currently be performed on English text (Zoho Corporation, 2026a).

Table 5.3: AI Component Implemented

Aspect	Description
Input	The text of the Course Feedback field
Processing	The native Zia Sentiment Analysis field of Zoho Creator (natural language processing performed by the platform)
Output	A sentiment label, Positive, Negative or Neutral, stored in the Sentiment field of the record
Analytics	Total and sentiment-wise counts, teacher-wise statistics, sentiment distribution charts and teacher-wise comparison charts on the administrator dashboard

5.10 Pro-Code Extension

No custom machine learning model was developed, trained or deployed, and no pro-code extension is described in this report. The requirements of this project, which are feedback collection, automatic sentiment labelling, statistics and reporting, are met by configuring built-in features of Zoho Creator.

5.11 Mobile and Portal Access

The application was exercised through a web browser (Google Chrome, Version 157.0) on a Windows 10/11 laptop. Mobile access and a separate portal are not covered in this report.

CHAPTER 6
TESTING

6.1 Testing Strategy

Each module is tested with valid and invalid input, and the separate Student and Faculty test users are used to verify form access restrictions. Sample comments that clearly express positive, negative and neutral opinions are submitted, and the Zia-generated sentiment is checked. Dashboard and report figures are compared with stored records. The RBAC cases in Table 6.1 record the confirmed access results.

6.2 Test Cases

Table 6.1 lists the test cases. The expected results describe the intended behaviour of the application.

Table 6.1: Test Cases and Results

TC ID	FR Ref.	Test Description	Expected Result	Status
TC-01	FR-01, FR-02	Open the Course Feedback form	Form shows Teacher Name, Course / Subject Name and Course Feedback; no sentiment dropdown	Pass
TC-02	FR-02	Try to enter a value in the Sentiment field	Student cannot enter or select a sentiment value	Pass
TC-03	FR-01	Submit the form with a required field empty	Record not saved; message asks for the missing field	Pass
TC-04	FR-03	Submit “The teacher explains every topic clearly and the classes are very helpful.”	Sentiment = Positive	Pass
TC-05	FR-03	Submit “The lectures are confusing and the teacher does not answer questions.”	Sentiment = Negative	Pass
TC-06	FR-03	Submit “The class is held twice a week in the morning.”	Sentiment = Neutral	Pass
TC-07	FR-07	Compare the dashboard total with the number of records	Both numbers are equal	Pass
TC-08	FR-07	Add the Positive, Negative and Neutral counts	Sum equals the total number of submissions	Pass
TC-09	FR-06, FR-09	Compare a teacher’s statistics with the filtered report	Counts match	Pass
TC-10	FR-07, FR-08	Submit new feedback and refresh the dashboard	Counts and charts include the new record	Pass
TC-11	FR-08	Open the teacher-wise comparison chart	Every teacher with feedback is shown	Pass
TC-12	FR-09	Filter the report by Sentiment = Negative	Only Negative records are listed	Pass

TC ID	FR Ref.	Test Description	Expected Result	Status
TC-13	FR-09	Filter the report by Teacher Name or Course / Subject Name	Only matching records are listed	Pass
TC-14	FR-10	Sign in as the Student test user and open Feedback Submission	Feedback Submission is accessible	Pass
TC-15	FR-10	As Student, attempt to open Feedback Review	Access is restricted	Pass
TC-16	FR-11	Sign in as the Faculty test user and open Feedback Review	Feedback Review is accessible	Pass
TC-17	FR-11	As Faculty, attempt to open Feedback Submission	Access is restricted	Pass
TC-18	FR-12	Sign in as Administrator and open administrative components	Administrative access is retained	Pass

6.3 Verification of the AI-Generated Sentiment

Because classification is performed by Zoho Creator's native Zia field, the AI-generated label is verified by behaviour. TC-04, TC-05 and TC-06 check representative positive, negative and neutral comments. TC-14 to TC-18 verify Student and Faculty form access restrictions and retained Administrator access. The remaining test cases check that dashboard counts, teacher-wise statistics, charts and filters agree with stored records.

6.4 Known Limitations

No defect log was kept for this project. The limitations below were identified from the Zoho Creator documentation and from the results; they are discussed further in Section 8.4.

Table 6.2: Known Limitations

S.No.	Limitation	Effect
1	Sentiment analysis currently works on English text (Zoho Corporation, 2026a).	Feedback in other languages is outside the scope of the application.
2	The label is an estimate made by the platform and is limited to Positive, Negative and Neutral.	The label shows the overall tone of a comment but not the reason for it.
3	The accuracy of the labels was not independently evaluated in this project.	The administrator still needs to read the comments behind the counts.

6.5 User Acceptance Testing

A user acceptance testing session with end users is not recorded in this report. The test cases in Table 6.1 were executed on the sample data described in Section 5.1.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is hosted on the Zoho Creator platform and was developed and tested using Google Chrome (Version 157.0) on a Windows 10/11 laptop with sample data of 5 teachers, 5 courses and 6 feedback records. This report does not record separate Development, Stage or Production environments, a publishing step or a release date, so none is claimed.

7.2 Performance Optimisation

No performance measurement was carried out and none is claimed. In the design, the classification is performed by the platform's built-in AI field when a record is saved, and the teacher-wise statistics, counts and charts read the stored Teacher Name and Sentiment fields directly, so no separate classification is performed when the dashboard is opened.

7.3 Security Features

Security controls include form-level separation through the Student and Faculty custom permission sets, tested restricted access, the disabled sentiment field and required-field validation. Table 7.1 summarizes the implemented controls. Encryption configuration, advanced authentication settings and audit controls are not described or claimed.

Table 7.1: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Disabled Sentiment field	Sentiment Analysis field of the Course Feedback form	A student entering or changing a sentiment value (TC-02)
No sentiment dropdown or rating selector	Course Feedback form	Sentiment depending on the student's choice instead of the words written
Required-field check	Course Feedback form	Incomplete records being saved (TC-03)
Field shown only to the administrator (platform option)	Sentiment Analysis field	Sentiment values being visible to other users (Zoho Corporation, 2026a)
Student custom permission set	Zoho Creator User Permissions	Restricts Feedback Review; Student test confirmed Feedback Submission access
Faculty custom permission set	Zoho Creator User Permissions	Restricts Feedback Submission; Faculty test confirmed Feedback Review access
Administrator access	Application permissions	Administrative access retained and tested

7.4 Governance and Administration

The administrator maintains the teacher and course entries, so that the same names are used in every feedback record and the teacher-wise statistics group correctly. The accuracy of the AI labels was not independently evaluated, and an evaluation against manual labels is proposed as future work (Section 9.2). Audit trail, backup and licence management are not described in this report and are not claimed.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario uses the sample records in the application, which contain 5 teachers, 5 courses and 6 feedback records. A student opens the Course Feedback form, selects the teacher and the course and enters the comment “Best guidance in lab and theory”. When the form is submitted, the Zia Sentiment Analysis field classifies the text and the record shows the sentiment Positive (Figures 8.1 and 8.2). The administrator dashboard then includes the record in its counts and charts (Figure 8.3). Table 8.1 gives the sentiment summary of the sample data.

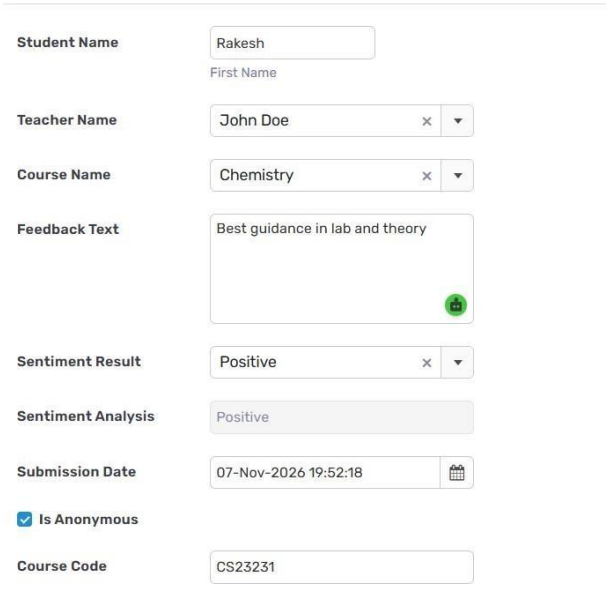


Figure 8.1: Scenario Step 1 – Student Submits Feedback (Course Feedback Form)

All Feedback Submissions

Student Name	Teacher Name	Course Name	Feedback Text	Sentiment Result	Submission Date	Feedb...	Is Anonymous	Course Cod
Robert Taylor	John Doe	Physics	Very helpful resources and supportive environment.	Neutral	07-Nov-2026 16:00:00	5	true	BIO201
Maria Garcia	John Doe	English	The assignment was too difficult for our level.	Neutral	07-Nov-2026 09:20:00	4	false	HIST150
James Wilson	Jane Smith	English	Excellent teaching style and clear explanations.	Positive	07-Nov-2026 14:45:00	3	true	ENG302
Sarah Lee	Maria Garcia	Chemistry	Great content but the slides were hard to read.	Positive	07-Nov-2026 11:15:00	2	false	MATH205
Alex Morgan	Jane Smith	History	The lecture was engaging and informative.	Neutral	07-Nov-2026 10:30:00	1	true	CS101
qwert	John Doe	English	good guidance in both theory and laboratory	Positive	07-Nov-2026 18:58:41	6	true	wefwgf

Figure 8.2: Scenario Step 2 – Feedback Records with AI-Generated Sentiment

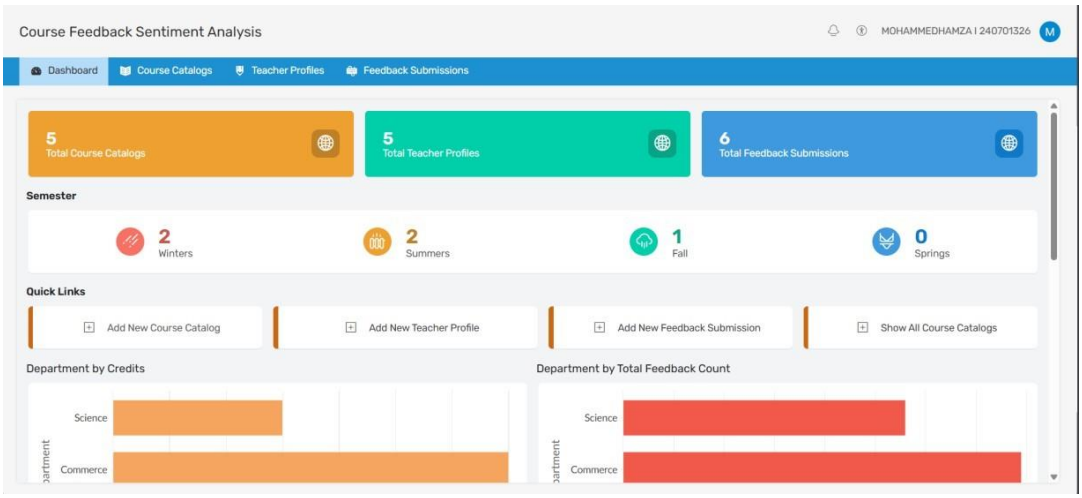


Figure 8.3: Scenario Step 3 – Administrator Dashboard

Table 8.1: Sentiment Summary of the Sample Data

Measure	Count
Total feedback submissions	6
Positive feedback	3
Negative feedback	0
Neutral feedback	3

The design ensures that the sentiment is produced by the platform and not chosen by the student, so every record is labelled in the same way. The dashboard and reports turn the labelled records into counts, teacher-wise statistics and charts, and the filters lead back to the individual comments.

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Simple feedback form with Teacher Name, Course / Subject Name and Course Feedback	Sections 4.3, 5.2	Form with three student fields and no sentiment dropdown (Figure 5.1; TC-01)
Automatic classification with the Zia Sentiment Analysis field	Sections 4.4, 5.9	Sentiment generated automatically and not entered by the student (Figures 5.2, 5.3; TC-02, TC-04 to TC-06)
Teacher and course data maintained	Sections 4.2, 5.2	Entries managed by the administrator so that feedback is grouped consistently
Dashboard with counts, teacher-wise statistics and charts	Sections 4.5, 5.4	Dashboard and charts (Figures 5.4, 5.5; TC-07 to TC-11)
Filterable feedback reports	Section 5.3	Filters by Sentiment, Teacher Name and Course / Subject Name (TC-12, TC-13)

8.3 Comparison with the Existing System

Table 8.3 compares the existing manual process with the proposed application. The comparison is qualitative; the time taken to process feedback was not measured.

Table 8.3: Existing System versus Proposed System

Parameter	Existing System (typical manual feedback)	Proposed System
Collecting feedback	Paper forms or general-purpose online forms	One Zoho Creator form with three fields
Sentiment of a comment	Judged by reading, or chosen by the student from a list	Generated automatically by the Zia Sentiment Analysis field; not chosen by the student
Consistency	Depends on the person reading the comment	The same built-in AI feature labels every record
Teacher-wise summary	Compiled by hand	Teacher-wise positive, negative and neutral statistics
Charts	Built manually, if at all	Sentiment distribution and teacher-wise comparison charts
Reports and filtering	Manual sorting of records	Feedback reports with filters

8.4 Limitations of the Low-Code Approach

The approach has limitations. According to the Zoho Creator documentation, sentiment analysis currently works on English text, so feedback in other languages is outside the scope of the application (Zoho Corporation, 2026a). The label is an estimate made by the platform and is limited to Positive, Negative and Neutral, and the accuracy of the labels was not independently evaluated in this project. The label shows the overall tone of a comment but not the reason for it, so the administrator still needs to read the comments behind the counts. Because a ready-made field is used, the labelling follows the behaviour of the platform's algorithm and cannot be tuned within this project.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is configuration: the form and its fields, the Sentiment Analysis field, the reports with filters and the dashboard. The artificial intelligence is added by configuring a built-in AI field, not by writing code. No Deluge script is used for the classification, and no custom machine learning model was developed, trained or deployed. Developing such a model would be pro-code work and is outside the scope of this project. For a feedback application of this size, configuring the built-in features was enough.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project developed the Course Feedback Sentiment Analysis System in Zoho Creator. It supports student feedback submission, faculty feedback review and resolution, faculty and course management, reports and dashboard monitoring. Zoho Creator's native Zia Sentiment Analysis classifies feedback as Positive, Negative or Neutral. Student and Faculty custom permission sets, configured through User Permissions, separate access to the Feedback Submission and Feedback Review forms; their restrictions were confirmed with separate test users. The Administrator retains administrative access. The result is a more structured way to collect, review and monitor course feedback without a custom sentiment model or external AI service.

The project shows that a useful AI-assisted application can be assembled on a low-code platform by configuring a built-in AI feature instead of training a model. Its main limitations are that the AI label is an estimate limited to three values, that the field currently supports English text, and that the accuracy of the labels was not independently evaluated.

9.2 Future Enhancements

- Evaluate the accuracy of the AI labels by comparing them with manual labels on a sample of comments.
- Add the keyword extraction AI field of Zoho Creator to show the topics that students mention most often.
- Add semester-wise or date-wise trend charts to show how the sentiment for each teacher changes over time.
- Send an email alert to the administrator when a Negative feedback record is submitted.
- Add a numeric rating beside the written comment, and support other languages when the platform supports them.

9.3 Learning Outcomes

Through this project the student built feedback submission and review components, faculty and course management, reports and a dashboard; configured Zoho Creator's native Zia AI field; and configured and tested Student and Faculty custom permission sets through User Permissions. The project demonstrates how low-code configuration can support feedback analysis, role-based access and monitoring while retaining the platform's limitations:

the label is an estimate, it supports English text, and its accuracy has to be evaluated separately.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Only features that are described in this report or visible in its screenshots are marked Y. Features that are not used or not reported are marked N and are not claimed.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and menu organised	Y	Course Feedback Sentiment Analysis; tabs Dashboard, Course Catalogs, Teacher Profiles, Feedback Submissions	4.5	Fig. 5.4
II	Form with several field types	Y	Course Feedback form: dropdown, multi-line and Sentiment Analysis (Zia AI) fields	4.3, 5.2	Fig. 5.1
II	Dropdown fields for Teacher Name and Course / Subject Name	Y	Course Feedback form	4.2, 4.3	Fig. 4.2
II	Lookup field / subform	N	Not reported	–	–
II	Field validations / layout rules	Partial	Required-field message tested (TC-03); no other validation reported	6.2	Table 6.1
II	List report with filter	Y	Feedback report filtered by Sentiment, Teacher Name, Course / Subject Name	5.3	Fig. 5.3
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	–
II	Chart report / dashboard charts	Y	Sentiment distribution and teacher-wise comparison charts	5.4	Figs. 5.4, 5.5
II	Page / dashboard built with UI Builder	Y	Administrator dashboard	5.4	Figs. 5.4, 5.5
II	Deluge – form events, functions, fetch, loops	N	No Deluge script used for the classification	5.5	–
II	Form workflows	Y	Zoho Creator form workflows support submission and review; event details are not specified	5.6	Form workflow configuration
II	Email / in-app notifications	N	Not reported (email alert listed as future work)	9.2	–
II	Testing evidence	Y	13 test cases (TC-01 to TC-13)	6.2	Table 6.1
III	Zia AI field	Y	Sentiment Analysis field, source field: Course Feedback	5.9	Fig. 5.2
III	Integration with a Zoho service or external system	N	Out of scope	5.7	–
III	Connection / REST API call / webhook	N	Not used	5.7	–

II	Schedule / approval / Blueprint / batch workflows	N	Not reported or claimed	5.6	–
Unit	**Zoho Creator Feature**	**Done**	**Where in the Application**	**Report Section**	**Figure / Code No.**
III	Users and custom permission sets	Y	Student and Faculty custom permission sets assigned to separate test users through User Permissions; Administrator access retained	4.6, 5.8, 6.2	Tables 4.5, 6.1
III	Analytics (charts and counts)	Y	Dashboard counts, teacher-wise statistics and charts	5.9	Figs. 5.4, 5.5
IV	Performance optimisation applied and measured	N	Not measured	7.2	–
IV	Deployment across environments	N	Not reported	7.1	–
IV	Security controls	Partial	Disabled Sentiment field; no sentiment dropdown; required-field check	7.3	Table 7.1
IV	Governance (audit trail, administration)	N	Not reported	7.4	–
V	Pro-code extension / custom ML model	N	Not used; no custom model was developed or trained	5.10	–
V	Low-code limitations and spectrum analysis	Y	Written discussion	8.4, 8.5	–
—	Mobile app / portal access	N	Not covered	5.11	–

Total features confirmed as implemented: 12 / 25 (a further 2 are partly implemented).

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2, 4.1		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5, 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7, 5.8 (no integration in scope)		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 1.5, 5.9, 5.10, 8.4, 8.5, 9		

The levels are to be entered by the student and the examiner. The application uses a built-in Zia AI field and no hand-written script, integration or production deployment is reported, as stated in Chapters 5 and 7.

APPENDIX C – SENTIMENT ANALYSIS FIELD CONFIGURATION

The application does not contain hand-written Deluge source code. The sentiment is generated by the built-in Sentiment Analysis field. The listing below records the configuration and the test inputs, so that the logic can be rebuilt or reviewed.

C.1 Sentiment Analysis Field of the Course Feedback Form

```
// Form         : Course Feedback
// Field type   : Sentiment Analysis (Zia AI field)
// Source field : Course Feedback (multi-line, same form)
// Output       : Positive | Negative | Neutral
// Behaviour    : appears disabled on the form; the student cannot
//                enter a value; value is estimated by the platform
// Language     : English text (current platform support)
// Custom model, keyword rules, Deluge script : none
```

C.2 Sample Test Comments (Table 6.1)

```
TC-04 : "The teacher explains every topic clearly and the classes are
very helpful."
        expected Sentiment = Positive
TC-05 : "The lectures are confusing and the teacher does not answer
questions."
        expected Sentiment = Negative
TC-06 : "The class is held twice a week in the morning."
        expected Sentiment = Neutral
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open a web browser (the application was tested in Google Chrome, Version 157.0).
2. Open the Course Feedback Sentiment Analysis application on Zoho Creator.
3. The menu shows the Dashboard, Course Catalogs, Teacher Profiles and Feedback Submissions tabs.

D.2 Instructions for the Student

1. Open the Course Feedback form.
2. Select the Teacher Name and the Course / Subject Name.
3. Type the feedback in the Course Feedback box.
4. Click Submit. Do not choose a sentiment; the Sentiment field is disabled and is filled in automatically as Positive, Negative or Neutral.

D.3 Instructions for Faculty and Administrator

1. Faculty users sign in with the assigned Faculty permission set and open Feedback Review to review and resolve feedback. Access to Feedback Submission is restricted for this permission set.
2. The Administrator retains administrative access, maintains permitted application data, and uses the dashboard and reports for monitoring.
3. To read the comments behind the figures, open the feedback report and filter it by Sentiment, Teacher Name or Course / Subject Name.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	I. Mohammed Hamza
Register No.	2116240701326
Project Title	Course Feedback Sentiment Analysis using Zoho Creator
Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Signature of External Examiner

Name: _____ *Name:* _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	I. Mohammed Hamza
Register No.	2116240701326
Department / Section	
Date of Viva	
Application URL Verified	Yes / No
Demo Duration	____*m*

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER EXTERNAL EXAMINER

Name: _____ *Name:* _____